Water Jug Problem

AIM: To write a Python program to solve the Water Jug Problem using state-space search.

Algorithm:

1. Start the program.
2. Initialize the capacities of the two jugs and the target quantity.
3. Represent the current state using the water quantities in both jugs.
4. Generate possible actions: fill a jug, empty a jug, or pour water from one jug into the other.
5. Apply the actions to generate new states.
6. Continue searching until the target quantity is obtained.
7. Display the sequence of states leading to the solution.
8. Stop the program.

Code :

exp03_water_jug.py > ...

```python
from collections import deque

def water_jug_problem(cap1, cap2, target):
    visited = set()
    queue = deque()
    queue.append(((0, 0), []))
    visited.add((0, 0))

    while queue:
        (j1, j2), path = queue.popleft()

        if j1 == target or j2 == target:
            return path + [(j1, j2)]

        next_states = [
            ((cap1, j2), "Fill Jug 1"),
            ((j1, cap2), "Fill Jug 2"),
            ((0, j2), "Empty Jug 1"),
            ((j1, 0), "Empty Jug 2"),
            ((j1 - min(j1, cap2 - j2), j2 + min(j1, cap2 - j2)), "Pour Jug 1 -> Jug 2"),
            ((j1 + min(j2, cap1 - j1), j2 - min(j2, cap1 - j1)), "Pour Jug 2 -> Jug 1")
        ]

        for state, action in next_states:
            if state not in visited:
                visited.add(state)
                queue.append((state, path + [(j1, j2, action)]))

    return None

cap1, cap2, target = 4, 3, 2
solution = water_jug_problem(cap1, cap2, target)

if solution:
    print(f"Steps to measure {target} liters using jugs of capacity {cap1} and {cap2}:\n")
    for step in solution[:-1]:
        print(f"Jug1: {step[0]}, Jug2: {step[1]}  ->  Action: {step[2]}")
```

Output :

```
PS D:\CSA1702-AI\Experiments> & C:\python311\python.exe d:/CSA1702-AI/Experiments/exp03_water_jug.py
Steps to measure 2 liters using jugs of capacity 4 and 3:

Jug1: 0, Jug2: 0  ->  Action: Fill Jug 2
Jug1: 0, Jug2: 3  ->  Action: Pour Jug 2 -> Jug 1
Jug1: 3, Jug2: 0  ->  Action: Fill Jug 2
Jug1: 3, Jug2: 3  ->  Action: Pour Jug 2 -> Jug 1
Jug1: 4, Jug2: 2  ->  Target Reached!
PS D:\CSA1702-AI\Experiments>
```

Result :

Thus, the Python program to solve the Water Jug Problem was executed successfully.